

Spatial Degradation of Hepatocyte Zonation — Signature-Set Battery

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Paper 1 (disease): Gribben/Vallier 2024, snRNA-seq, 47 donors, ~69k hepatocytes (GSE202379).
Paper 2 (healthy ruler): Yakubovsky/Itzkovitz 2026 spatial atlas; $\eta = \frac{\text{sumPP}}{\text{sumPP} + \text{sumPC}}$.

METHOD: build candidate PC/PP gene sets; score each as $\text{coord} = \text{mean}_z(\text{PC}) - \text{mean}_z(\text{PP})$;
judge RULER QUALITY on HEALTHY metrics only (validation signs, PC-PP anticorrelation,
split-half reproducibility). Disease H1/H2/H3 are TRANSFER results, never used to pick a set.

H1 = donor-level collapse of zonation across stage (spread down, PC-PP anticorr $\rightarrow 0$).

H2 = zonal-slope loss of zonation genes (held-out, donor-level).

H3 = de-zonation $\leftrightarrow$ plasticity coupling (donor-level).

See: candidate_set_build_report.csv, signature_battery_summary.csv, selected_set_decision.txt.

Signature battery — ruler quality (HEALTHY) + recommended role

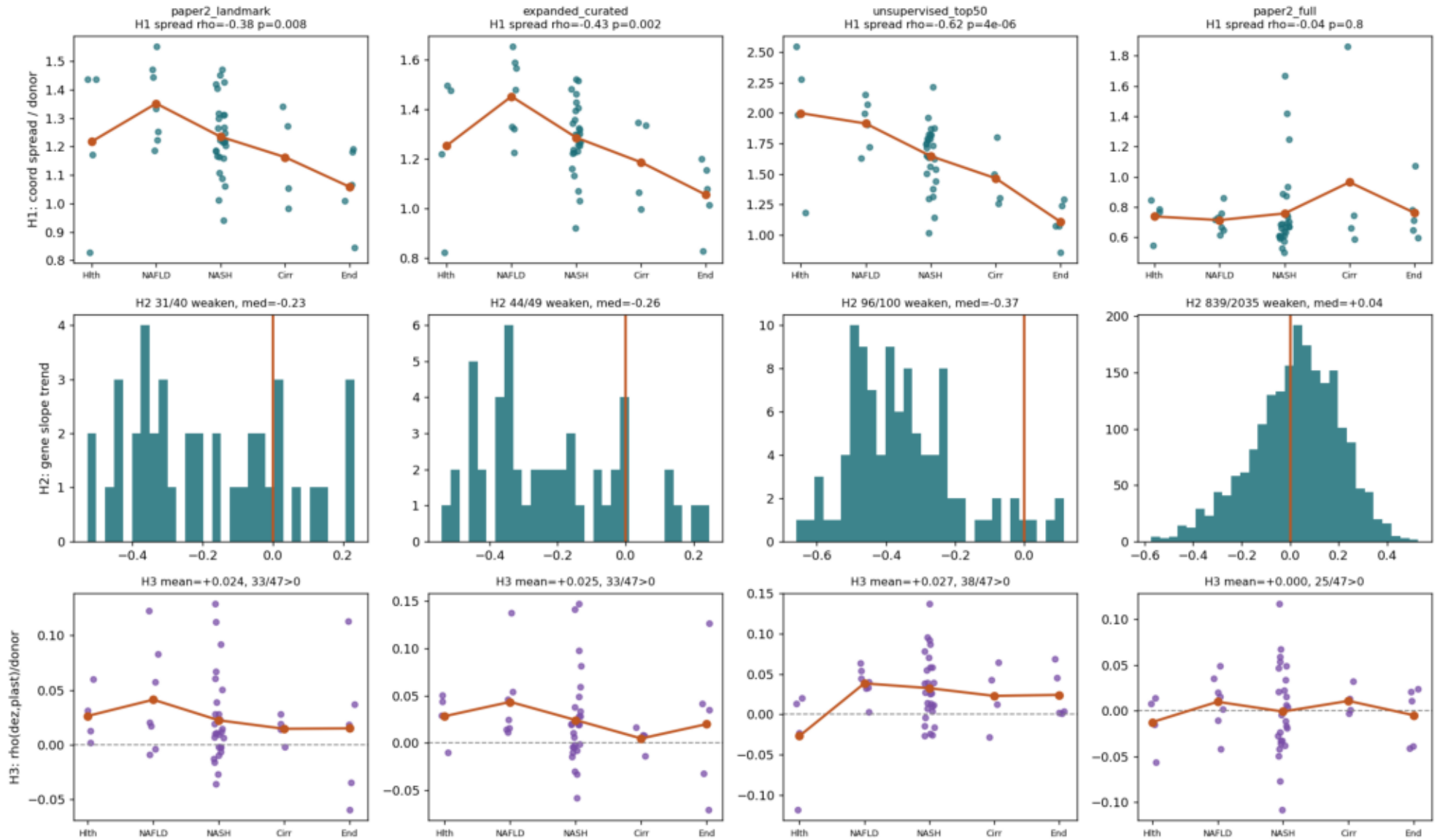
set_name	n_pc	n_pp	validation_pass	healthy_pc_pp_anticorr	healthy_splithalf_rho_mea	recommended_role
paper2_landmark	20	20	True	-0.455	0.631	primary_candidate
core_curated	13	8	True	-0.278	0.537	interpretability_only
expanded_curated	26	23	True	-0.479	0.67	primary_candidate
paper2_top50	50	50	True	-0.404	0.573	primary_candidate
paper2_top100	100	100	True	-0.273	0.563	primary_candidate
paper2_top250	250	250	True	0.0408	0.554	exploratory
paper2_full	1624	447	True	0.556	0.484	exploratory
selected_frozen	26	23	True	-0.479	0.67	primary_candidate
full	0	0	True	0.532	0.464	exploratory
supervised	0	0	True	-0.00504	0.452	exploratory
unsupervised	0	0	True	-0.306	0.781	interpretability_only
unsupervised_top100	100	100	True	-0.292	0.821	primary_candidate
unsupervised_top50	50	50	True	-0.306	0.785	primary_candidate

Disease TRANSFER results (H1/H2) — NOT used for set selection

set_name	h1_coord_spread_rho_vs_std	h1_coord_spread_perm1	h1_pp_anticorr_rho_vs_std	h2_frac_weakening	h2_median_trend_rho	recommended_role
paper2_landmark	-0.38	0.0085	0.608	0.775	-0.229	primary_candidate
core_curated	-0.541	0.0005	0.562	1	-0.339	interpretability_only
expanded_curated	-0.434	0.003	0.612	0.898	-0.258	primary_candidate
paper2_top50	-0.316	0.0355	0.564	0.727	-0.204	primary_candidate
paper2_top100	-0.284	0.0605	0.423	0.591	-0.0664	primary_candidate
paper2_top250	-0.265	0.067	0.25	0.53	-0.0112	exploratory
paper2_full	-0.0422	0.767	-0.0397	0.412	0.0369	exploratory
selected_frozen	-0.434	0.003	0.612	0.898	-0.258	primary_candidate
full	0.021	0.883	0.00651	0.415	0.0341	exploratory
supervised	0.122	0.403	0.569	0.87	-0.169	exploratory
unsupervised	-0.562	0.0005	0.607	0.96	-0.375	interpretability_only
unsupervised_top100	-0.608	0.0005	0.613	0.915	-0.324	primary_candidate
unsupervised_top50	-0.617	0.0005	0.607	0.96	-0.372	primary_candidate

Four rulers x H1/H2/H3 (donor-level)

Zonation-collapse battery: 4 rulers x H1/H2/H3 (donor-level)



H2b — program-level zonal-slope loss (paper2_full; more negative = faster)

program	n_genes	median_trend_rho	frac_weakening	n_q05_weakening
wnt_pericentral_identity	5	-0.436	1	5
urea_nitrogen	12	-0.285	0.583	7
cyp_xenobiotic	23	-0.237	0.783	16
lipid_metabolism	16	-0.034	0.5	8
other	1946	0.038	0.405	562
bile_acid_transport	9	0.0412	0.444	3
cholangiocyte_plasticity	2	0.0563	0.5	0
acute_phase_secreted	22	0.0904	0.318	6

Selected set decision + unsupervised agreement

SELECTED set: expanded_curated

Reason: best HEALTHY-ruler quality among primary candidates.

Healthy metrics used (NOT disease collapse):

- validation: 8/8 markers correct

- healthy PC-PP anticorrelation: -0.479 (want strongly negative)

- healthy split-half reproducibility: +0.670

- size: 26 PC + 23 PP

WARNING: selected using Paper 2 / healthy-ruler criteria ONLY, NOT Paper 1 disease effect (disease is the transfer/test cohort)

Unsupervised(PCA) vs landmark-signature agreement: $\rho(\text{axis}, \text{coord}) = +0.569$, $\rho(\text{axis}, \text{pc}) = +0.388$, $\rho(\text{axis}, \text{pp}) = -0.453$.